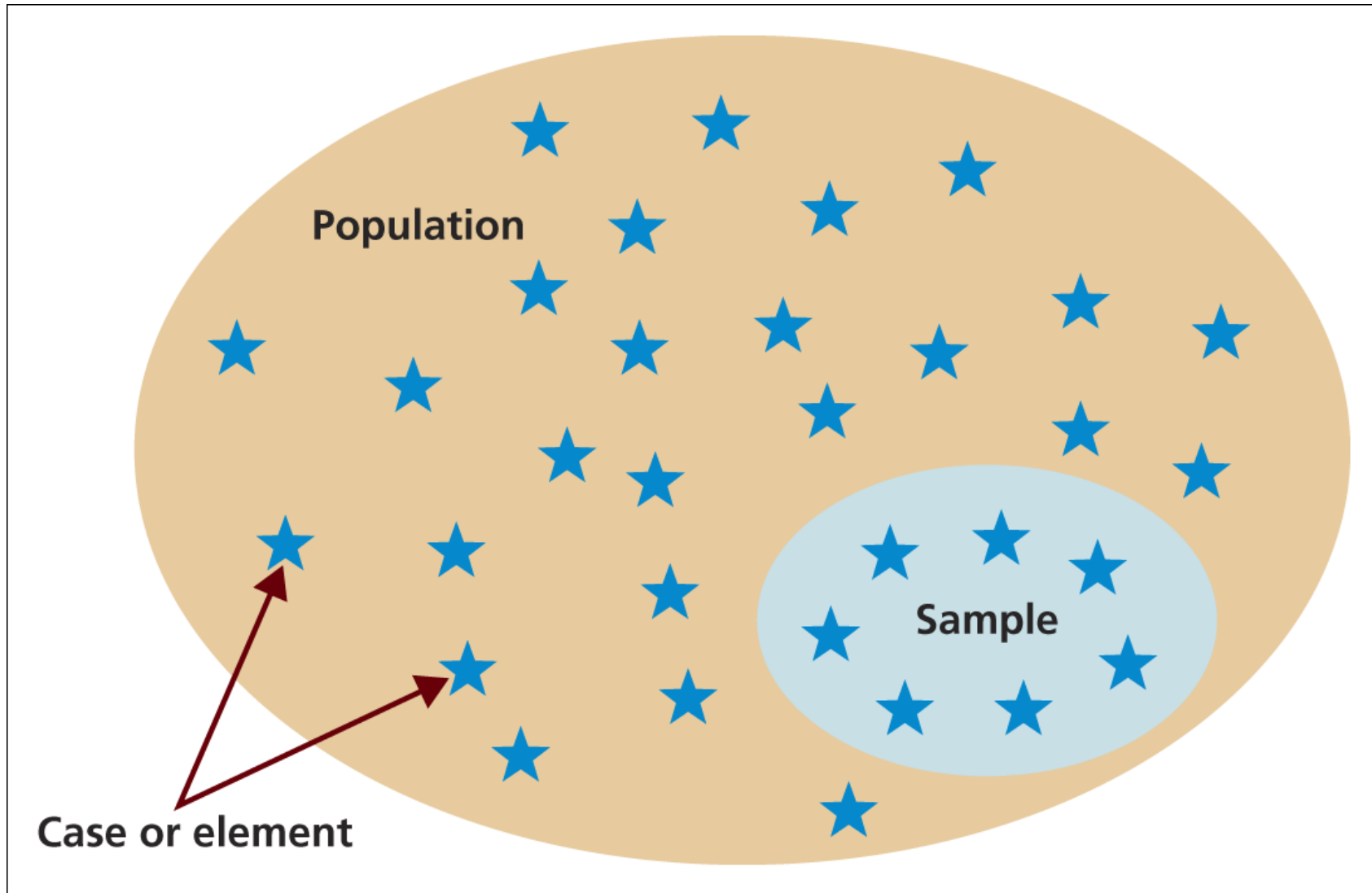


PSOT

Department of Electronics and Telecommunications

Population and Sample



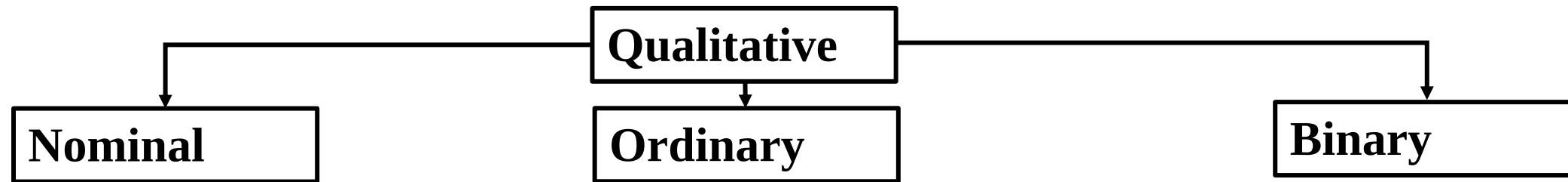
Data Types

- Data types: qualitative – nominal, ordinary, binary;
- Data types: quantitative – continuous, discrete
- Data types: cross-sectional, time series
- Data types: uni-variate, bi-variate, multi-variate

Data Types

- Data types: qualitative – nominal, ordinary, binary;

A **qualitative data types** are variables that are **not numerical** and which values fits into **categories or levels**



where no ordering is possible

For example:

Eye colour
Country
Gender
Types of cars

order implied in the levels.

For example:

severity of road accidents:
light, moderate, Fatal
Winning Medal
Gold, silver, bronze
Exam Ranking
Feedback scale 1-10

categorical data that can take exactly two possible values,

For example:

such as "A" and "B", or
"heads" and "tails"
True and False
Pass and fail

Data Types

- Data types: qualitative – nominal, ordinary, binary;

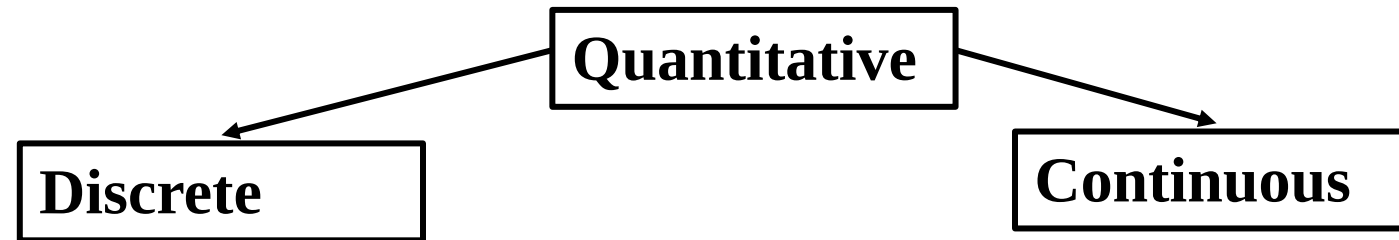
Difference between Nominal and Ordinal Data

Nominal Data	Ordinal Data
Nominal data can't be quantified, neither they have any intrinsic ordering	Ordinal data gives some kind of sequential order by their position on the scale
Nominal data is qualitative data or categorical data	Ordinal data is said to be "in-between" qualitative data and quantitative data
They don't provide any quantitative value, neither we can perform any arithmetical operation	They provide sequence and can assign numbers to ordinal data but cannot perform the arithmetical operation
Nominal data cannot be used to compare with one another	Ordinal data can help to compare one item with another by ranking or ordering
Examples: Eye colour, housing style, gender, hair colour, religion, marital status, ethnicity, etc	Examples: Economic status, customer satisfaction, education level, letter grades, etc

Data Types

- Data types: quantitative – continuous, discrete

A **quantitative** variable is a variable that reflects a notion of **magnitude**, that is, if the values it can take are **numbers**.



can take are **countable** and have a **finite number of possibilities**

For example:

Number of children per family,
Number of students in a class,
Number of citizens of a country
No. of cases for Covid-19

can take values are **not countable** and have an **infinite number of possibilities**.

For example:

Age, Weight, Height
Wi-Fi Frequency, Speed

	Time Series Data	Cross Sectional Data
DEFINITION	A type of data consisting of observations of a single subject at multiple time intervals.	A type of data consisting of observations of many subjects at the same point in time.
MAIN FOCUS	Focuses on the same variable over a period of time.	Focuses on several variables at the same point in time.
EXAMPLES	Profit of an organization over a period of 5 years' time	Maximum temperature of several cities on a single day

Time series Vs Cross-Sectional data

City	Date	MaxTemperature	Humidity	Wind
NYC	1/1/2012	35	56%	3 mph
NYC	1/1/2013	47	65%	21 mph
NYC	1/1/2014	30	39%	16 mph
NYC	1/1/2015	55	45%	4 mph

City	Date	MaxTemperature	Humidity	Wind
NYC	1/1/2015	55	45%	4 mph
SFO	1/1/2015	70	35%	21 mph
Boston	1/1/2015	34	39%	16 mph
Chicago	1/1/2015	29	15%	54 mph

Chapter 3: Data Preprocessing

- Data Preprocessing: An Overview
 - Data Quality
 - Major Tasks in Data Preprocessing
- Data Cleaning
- Data Integration
- Data Reduction
- Data Transformation

Data Quality: Why Preprocess the Data?

- Measures for data quality: A multidimensional view
 - Accuracy: correct or wrong, accurate or not
 - Completeness: not recorded, unavailable, ...
 - Consistency: some modified but some not, dangling, ...
 - Timeliness: timely update?
 - Believability: how trustable the data are correct?
 - Interpretability: how easily the data can be understood?

Major Tasks in Data Preprocessing

- **Data cleaning**
 - Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- **Data integration**
 - Integration of multiple databases, data cubes, or files
- **Data reduction**
 - Dimensionality reduction
 - Numerosity reduction
 - Data compression
- **Data transformation and data discretization**
 - Normalization
 - Concept hierarchy generation